📄 EduTrack — Project Proposal / Synopsis

Academic Management Platform

VTU Internship 2026 — Team 15

1. Introduction

In today's educational institutions, managing academic activities such as attendance, assignments, marks, and course materials is often done manually or through disconnected systems. This leads to inefficiencies, data inconsistencies, and poor communication between students, teachers, and administrators.

EduTrack is a web-based academic management platform designed to digitize and centralize all academic operations. It provides a unified platform where students can track their academic progress, teachers can manage their courses and students, and administrators can oversee the entire institution — all in real time through a secure, role-based web application.

2. Problem Statement

Educational institutions face several challenges in managing academic data:

- Attendance is marked manually on paper, making it difficult to track and analyze
- Marks and grades are maintained in spreadsheets with no centralized access
- Students have no real-time visibility into their academic performance
- Assignment deadlines and submissions are managed informally
- Administrators have no centralized tool to manage students, teachers, and platform settings
- Password recovery is insecure or non-existent in most systems
- Bulk data entry (adding multiple students or teachers) is time-consuming

EduTrack addresses all of these problems through a single integrated platform.

3. Objectives

- Build a full-stack web application for academic management
- Provide role-based access for Admin, Teacher, and Student
- Enable real-time attendance tracking with percentage calculation
- Allow teachers to enter marks with automatic grade calculation
- Support assignment creation, submission, and grading
- Enable course material upload and download
- Implement secure OTP-based password reset via email
- Allow bulk import of students, teachers, attendance, and marks via CSV
- Provide admin control over platform settings including maintenance mode
- Deploy the application on cloud platforms for public access

4. Proposed System

EduTrack is a three-tier web application consisting of:

Frontend

A React 19 Single Page Application (SPA) that provides separate portals for Admin, Teacher, and Student. The frontend communicates with the backend through REST APIs and supports dark/light theme switching.

Backend

A Spring Boot 3 REST API built with Java 21. It handles authentication, business logic, OTP generation and validation, email sending, file uploads, and CSV bulk imports. Passwords are secured using BCrypt hashing.

Database

A MySQL 8 relational database with 11 tables storing all academic data. Hibernate ORM manages the mapping between Java entities and database tables.

5. System Architecture

```
User (Browser)
      ↓
React Frontend (Vercel)
      ↓  REST API calls
Spring Boot Backend (Render)
      ↓  JDBC
```

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MySQL Database (Railway)
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6. Key Features

Admin Module

- Manage students and teachers (add, edit, delete, search)
- Bulk import via CSV
- Enable maintenance mode
- Control student registration
- Set active academic year and semester

Teacher Module

- Create and manage courses
- Mark and view attendance
- Enter and update student marks
- Create assignments and view submissions
- Upload course materials

Student Module

- View attendance with percentage
- View marks and grades
- View and submit assignments
- Download course materials
- Track academic progress on dashboard

Security

- BCrypt password hashing
- OTP-based password reset via Gmail SMTP
- Role-based access control
- Maintenance mode to block logins

7. Technology Stack

LAYER	TECHNOLOGY
Frontend	React 19, JavaScript
Backend	Spring Boot 3.2.3, Java 21
Database	MySQL 8.0
ORM	Hibernate (Spring Data JPA)
Security	Spring Security, BCrypt
Email	Gmail SMTP (Spring Mail)
Testing	Playwright (E2E), JUnit 5, Mockito
Deployment	Vercel, Render, Railway
Version Control	Git, GitHub

8. Database Design

The database consists of 11 tables:

TABLE	PURPOSE
admins	Admin accounts
teachers	Teacher accounts
students	Student accounts
courses	Course information
attendance	Daily attendance records
assignments	Course assignments
submissions	Student submissions
marks	Student marks and grades
course_materials	Study materials
otp_verification	OTP for password reset
system_settings	Platform configuration

9. Testing

TYPE	TOOL	COUNT
End-to-End Tests	Playwright	43 tests
Unit Tests	JUnit 5 + Mockito	31 tests
API Tests	Postman	26 requests

10. Deployment

COMPONENT	PLATFORM	URL
Frontend	Vercel	https://vtu-intern-2026-team15-java-fs.vercel.app
Backend	Render (Docker)	https://edutrack-backend-o1ym.onrender.com
Database	Railway (MySQL)	Cloud hosted

11. Expected Outcomes

- A fully functional academic management platform accessible via public URL
- Secure role-based access for Admin, Teacher, and Student
- Real-time attendance and marks tracking
- OTP-based password recovery for all users
- Automated grading system
- Bulk data import capability
- Comprehensive test suite with 100 tests across all layers
- Complete documentation including SDD, API docs, WBS, and user stories

12. Team

NAME	ROLE
Naveen P R	Team Leader & Core Developer
Rashmi S P	Developer
Sneha H S	Developer
Soujanya M	Developer

NAME	ROLE
Deepa G R	Developer
Kavana N	Developer
Vanashree R	Developer
Deeksha D R	Developer
Saud Shaik	Developer

13. Conclusion

EduTrack provides a comprehensive solution to the challenges faced by educational institutions in managing academic data. By centralizing attendance, marks, assignments, and course materials in a single platform with role-based access, EduTrack improves efficiency, transparency, and communication between all stakeholders. The platform is built using industry-standard technologies, follows best practices in security and testing, and is deployed on cloud infrastructure for public access.